

Brexit Uncertainty and Trade Disintegration

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A Online Appendix

A.1 Value Functions

The following derivation is adapted from Carballo et al. (2018). The export investment decision is ixV specific for each firm so we omit these subscripts without loss of generality. The expected value of starting to export at time t conditional on observing current conditions a_t , the firm's own unit costs c , and the demand regime r , is:

$$\Pi_e(a_t, c, r) = \pi(a_t, c) + \underbrace{\beta[(1 - \gamma)\Pi_e(a_t, c, r)]}_{\text{No Shock}} + \underbrace{\gamma\mathbb{E}\Pi_e(a', c, r)}_{\text{Shock}}. \quad (\text{A.1})$$

This includes current operating profits upon entering and the discounted future value. Without a shock the firm value next period remains $\Pi_e(a_t, c, r)$. If a shock arrives then a new a' is drawn, so the third term is the ex-ante expected value of exporting following a shock, $\mathbb{E}\Pi_e(a', c, r) = \mathbb{E}\pi(a', c)/(1 - \beta)$, where $\mathbb{E}$ denotes the expectation operator.

The expected value of waiting is

$$\Pi_w(c, r) = 0 + \underbrace{\beta(1 - \gamma + \gamma\bar{H}(a_t))\Pi_w(c, r)}_{\text{Wait}} + \underbrace{\beta\gamma(1 - \bar{H}(a_t))(\mathbb{E}\Pi_e(a' \geq \bar{a}, c, r) - K)}_{\text{Enter}}. \quad (\text{A.2})$$

A non-exporter at t receives zero profits from that activity today. The continuation value remains at Π_w if either demand is unchanged, with probability $1 - \gamma$, or changes to some level that is not sufficiently high to induce entry, with probability $\gamma\bar{H}(a_t)$. If demand changes and is above some endogenous trigger level, $a' \geq \bar{a}$, then we obtain the third term, reflecting the expected value of exporting net of the sunk cost, K , conditional on the new demand being high enough to trigger entry. The conditional expected value of exporting if $a' \geq \bar{a}$ is given by

$$\mathbb{E}\Pi_e(a' \geq \bar{a}, c, r) = \mathbb{E}\pi(a' \geq \bar{a}, c, r) + \beta(1 - \gamma)\mathbb{E}\Pi_e(a' \geq \bar{a}, c, r) + \beta\gamma\mathbb{E}\Pi_e(a', c, r). \quad (\text{A.3})$$

A firm with costs c_v is indifferent between entering or waiting if demand is at a threshold level $a_{c_v} = \bar{a}(c_v)$. Instead, of solving for $\bar{a}(c_v)$ we characterize the marginal exporting firm at any current demand, which is characterized by a cost parameter c_t^U defined by $a_t = \bar{a}(c_t^U)$.¹ If a firm has costs equal to this threshold, then in that period all other firms in that industry with lower costs also export to that particular destination.

We obtain an expression for this cutoff by using the entry condition in (2); the value functions in (A.1), (A.2) and (A.3), and the expression for $\mathbb{E}\Pi_e(a', c, r)$.

¹We can do so since a is common to all firms exporting to a given market in a given industry and the marginal cost is the only source of heterogeneity among such firms. Assuming a continuum of firms in any given industry with productivity that can be ranked according to a strictly increasing CDF, we can find the marginal export entrant for any a_t .

A.2 Industry Export Dynamics

We generalize the export value equation in (5) to allow for non-stationary periods as follows. Consider starting from a stationary period with a cutoff c_{i0V}^U followed by a single permanent shock to c_{i1V}^U observed at the end of period $t = 0$ and a constant a_{iV} . The constant a_{iV} would prevail if uncertainty increased but current conditions were unchanged. In this case, total exports can be written as the sum of: (i) exports by the fraction $1 - \beta^t$ of firms whose export capital was destroyed but have costs below the new cutoff c_{i1V} and thus re-invest; and (ii) export values given by equation (5) at the previous cutoff c_{i0V} multiplied by the survival probability β^t :

$$R(a_{iV}, c_{itV}^U, \beta^t) = \begin{cases} R(a_{iV}, c_{i1V}^U), & \text{if } c_{i1V}^U \geq c_{i0V}^U \\ [R(a_{iV}, c_{i1V}^U)(1 - \beta^t)] + [R(a_{iV}, c_{i0V}^U)\beta^t], & \text{if } c_{i1V}^U < c_{i0V}^U. \end{cases} \quad (\text{A.4})$$

Thus the estimation must account for lags of negative uncertainty shocks since they work via attrition. Each of these expressions applies to more general cases and allows for any history of shocks between $t = 0$ and $t - 1$ provided that either $c_{itV}^U \geq c_{i0V}^U$, so the first line in equation (A.4) applies; or $c_{itV}^U \in [c_{iT-1V}^U, c_{i0V}^U)$ in the second line of (A.4) applies. So the history we need to consider empirically is not necessarily of all shocks since c_{i0V}^U . We denote this potential dependence of exports on past cutoffs by the vector $\mathbf{c}_{itV}^U = \{c_{i0V}^U, \dots, c_{itV}^U\}$.

To derive the estimating equation (7) when we do not start in a stationary period we start by approximating each of the terms in $\square$ in the second line of equation (A.4). The expression in (6) shows how to approximate the stationary components in each t . However, we must also account for the fact that the relative weights on R_{ixVt} and R_{ixVt-T} depend on when the cutoffs changed, which may differ across destinations, i . We denote the dependence of those weights on prior shocks in i by the history coefficient, b_{it}^h , and approximate it around $\bar{b}_i^h$: interpreted as the average export death rate into i . This coefficient is equal to $1 - \beta^T$ if conditions have worsened in i for T periods before t , and 1 otherwise.² Thus the more general decomposition of (A.4) is

$$\ln \frac{R_{ixVt}}{R(\theta_{ixV})} = \left(k_c \hat{U}_{ixVt} + k_a \hat{a}_{ixVt} + \hat{N}_{xtV} \right) \bar{b}_i^h + \left(k_c \hat{U}_{ixVt-T} + k_a \hat{a}_{ixVt-T} + \hat{N}_{xt-TV} \right) (1 - \bar{b}_i^h) + o_{ixVt}. \quad (\text{A.5})$$

The first term in () in equation (A.5) is the same as in (6) after we use the definitions of c^U , c^D from (3) and define $k_a \equiv 1 + \frac{k_c}{\sigma-1}$. The second term in () is the approximation of the stationary value in $t - T$. The average export death rate, $\bar{b}_i^h$, provides the relative weight and the history coefficient b_{it}^h has no first order effects since R_t and R_{t-T} are approximated around common values. Applying identifying assumptions A1-A4 we obtain (7) where

$$\alpha_{ixV} \equiv \ln R(a_{ixV}, c_{ixV}^D) + (1 - \bar{b}_i^h) \tilde{k}_c \ln U_{ixV} \quad ; \quad \alpha_{it} \equiv \bar{b}_i^h \left[\tilde{k}_a \ln \frac{a_{ixtV}}{a_{ixV}} + n_t \right] + (1 - \bar{b}_i^h) \left[\tilde{k}_a \ln \frac{a_{ixt-TV}}{a_{ixV}} + n_{t-T} \right]$$

The fixed effect α_{ixV} controls for the deterministic state exports in a stationary state and the pre-sample uncertainty under A4; whereas α_{it} is obtained by using the definition of a , A1 and A3.

²We can allow for a more general history coefficient, b_{ixt}^h , that reflects bilateral variation in the history coefficient but the approximation would still be similar.

A.3 Uncertainty Factor Derivation

To obtain (14) we start with (13) and take a second order approximation with respect to $\mathbf{u} = (\omega^{BR}, \ln m)$ around $\mathbf{u}_0 = (1, \ln m_0)$. We treat the observed policy risk measures as the true beliefs. The general form of the approximation is

$$\ln U(\mathbf{u}) = \ln U(\mathbf{u}_0) + (\mathbf{u} - \mathbf{u}_0) \cdot \nabla \ln U(\mathbf{u}_0) + \frac{1}{2} (\mathbf{u} - \mathbf{u}_0)^T (\mathbf{H} \ln U(\mathbf{u}_0)) (\mathbf{u} - \mathbf{u}_0) + r,$$

where ∇ is the gradient function and $\mathbf{H} \ln U(\mathbf{u}_0)$ is the Hessian matrix and r the approximation error. Around no tail policy risk the following terms are zero: $\ln U(\mathbf{u}_0)$, the first and second derivatives with respect to $\ln m$. The tail risk derivatives and cross effects evaluated at $\mathbf{u}_0$ are

$$\begin{aligned} (\sigma - 1) \frac{\partial \ln U}{\partial \omega^{BR}}|_{\mathbf{u}_0} &= \frac{\partial \ln(1 + \tilde{\beta}m(\omega - 1))}{\partial \omega^{BR}}|_{\mathbf{u}_0} = \tilde{\beta}m_0 \\ (\sigma - 1) \frac{\partial^2 \ln U}{\partial (\omega^{BR})^2}|_{\mathbf{u}_0} &= \frac{\partial^2 \ln(1 + \tilde{\beta}m(\omega - 1))}{\partial (\omega^{BR})^2}|_{\mathbf{u}_0} = -(\tilde{\beta}m_0)^2 \\ (\sigma - 1) \frac{\partial^2 \ln U}{\partial \ln m \partial \omega}|_{\mathbf{u}_0} &= \frac{\partial^2 \ln(1 + \tilde{\beta}m(\omega - 1))}{\partial \ln m \partial \omega^{BR}}|_{\mathbf{u}_0} = \tilde{\beta}m_0 \\ \ln U(\mathbf{u}) &= \frac{\tilde{\beta}m_0}{\sigma - 1} (\omega^{BR} - 1) \left[1 - \tilde{\beta}m_0 \frac{(\omega^{BR} - 1)}{2} + \ln \frac{m}{m_0} \right] + r \end{aligned} \quad (\text{A.6})$$

where the first two terms in $[]$ represent the first and second order effects of ω^{BR} and the third represents the cross effect.

Re-introducing the subscripts in (A.6) that indicate the level of variation and replacing $\ln \frac{m_t}{m_0} = \Pr(\widehat{\mathcal{R}_{\mathcal{T}}|\mathcal{I}_t})$ with eq. (9) and using (10) to map to the estimation we have

$$\begin{aligned} \ln U_{ixVt} &= \frac{\tilde{\beta}_i m_{ix0}}{\sigma - 1} (\omega_{ixV}^{BR} - 1) \left[1 - \tilde{\beta}_i m_{ix0} \frac{(\omega_{ixV}^{BR} - 1)}{2} + \sum_l r_l^B \hat{B}_{t-l} + e_t^r \right] + r_{ixVt} \\ &= \frac{\tilde{\beta}_i m_{ix0}}{\sigma - 1} (\omega_{ixV}^{BR} - 1) \left[\sum_l r_l^B (mbv_{t-l} - mbv_0) + 1 - \tilde{\beta}_i m_{ix0} \frac{(\omega_{ixV}^{BR} - 1)}{2} + e_t^r \right] + r_{ixVt} \\ &= \frac{\tilde{\beta}_i m_{ix0}}{\sigma - 1} (\omega_{ixV}^{BR} - 1) \sum_l r_l^B mbv_{t-l} + \alpha_{ixV}^U + e_{ixVt}^r \\ &= \frac{\tilde{\beta}_i m_{ix0}}{\sigma - 1} \sum_{s=F, M, W} \eta_{ix}^s \left[(\tau_{ixV}^s)^{-\sigma} - 1 \right] \sum_l r_l^B mbv_{t-l} + \alpha_{ixV}^U + e_{ixVt}^r \\ &= \frac{\tilde{\beta}_i m_{ix0}}{\sigma - 1} \sum_{s=M, W} \eta_{ix}^s \left[(\tau_{ixV}^s)^{-\sigma} - 1 \right] \sum_l r_l^B mbv_{t-l} + \alpha_{ixV}^F + \alpha_{ixV}^U + e_{ixVt}^r. \end{aligned}$$

The second line replaces the information input $\hat{B}_{t-l}$ with $mbv_{t-l} - mbv_0$. The third line uses α_{ixV}^U and e_{ixVt}^r defined below. The fourth replaces ω_{ixV}^{BR} using (11) over all $s = F, M, W$ and the last one separates out the

bilateral-time effect from the FTA scenario and defines it as α_{ixt}^F , so we obtain (14) in the text.

$$\begin{aligned}\alpha_{ixt}^F &\equiv \frac{\tilde{\beta}_i m_{ix0}}{\sigma - 1} \eta_{ix}^F \left[(\tau_{ix}^F)^{-\sigma} - 1 \right] \sum_l r_l^B m b v_{t-l} \\ \alpha_{ixV}^U &\equiv \frac{\tilde{\beta}_i m_{ix0}}{\sigma - 1} (\omega_{ixV}^{BR} - 1) \left[\left(1 - \tilde{\beta}_i m_{ix0} \frac{(\omega_{ixV}^{BR} - 1)}{2} \right) - m b v_0 \sum_l r_l^B \right] \\ e_{ixVt}^r &\equiv r_{ixVt} + \frac{\tilde{\beta}_i m_{ix0}}{\sigma - 1} (\omega_{ixV}^{BR} - 1) e_t^r\end{aligned}$$

A.4 Export Estimation Equation Derivation

We obtain equation (15) by replacing the uncertainty factor in equation (7) with eq. (14). Below we show the steps to re-arrange and redefine the parameters to arrive at our estimating equation:

$$\ln R_{ixVt} = \bar{b}_i^h k_c \left[-\frac{\tilde{\beta}_i m_{ix0}}{\sigma - 1} \sum_{s=M,W} \eta_{ix}^s \sum_{l=0}^L r_l^B \left\{ m b v_{t-l} \left[1 - (\tau_{ixV}^s)^{-\sigma} \right] \right\} + \alpha_{ixt}^F + \alpha_{ixV}^U + e_{ixVt}^r \right] + \alpha_{ixV} + \alpha_{it} + o_{ixVt} \quad (\text{A.7})$$

$$= \sum_{s=M,W} \sum_{l=0}^L W_{ix}^s(l) \left\{ m b v_{t-l} \left[1 - (\tau_{ixV}^s)^{-\sigma} \right] \right\} + (\bar{b}_i^h k_c (\alpha_{ixt}^F + \alpha_{ixV}^U) + \alpha_{ixV}) + \alpha_{it} + (\bar{b}_i^h k_c e_{ixVt}^r + o_{ixVt}) \quad (\text{A.8})$$

$$= \sum_{s=M,W} \sum_{l=0}^L W_{ix}^s(l) \left\{ m b v_{t-l} \left[1 - (\tau_{ixV}^s)^{-\sigma} \right] \right\} + \alpha_{ixV,it,xt} + e_{ixVt} \quad (\text{A.9})$$

In the second line we rearrange terms and define $W_{ix}^s(l) \equiv -\bar{b}_i^h k_c \frac{\tilde{\beta}_i m_{ix0}}{\sigma - 1} \eta_{ix}^s r_l^B$. In the third, we collect terms into groups of estimated coefficients, a set of terms absorbed by fixed effects $\alpha_{ixV,it,xt}$, and the error term, where the latter two are given by:

$$\alpha_{ixV,it,xt} \equiv (\bar{b}_i^h k_c (\alpha_{ixt}^F + \alpha_{ixV}^U) + \alpha_{ixV}) + \alpha_{it}, \quad e_{ixVt} \equiv \bar{b}_i^h k_c e_{ixVt}^r + o_{ixVt}.$$

A.5 Product Export Dynamics

Similar to the value estimation we illustrate the results based on starting at some stationary state at $t - 12$ and then considering two alternative types of shock: a deterioration or improvement in business conditions.

We do not directly observe whether the unit cost of the most productive firm is below the cutoff, but we do observe whether a product is traded at each point in time. We use $\bar{r}_{ixVt}$ to denote the sales of the most productive firm in the cohort of firms in the cell xiV productivity distribution at a point in time. It need not identify the same firm across time periods due to churning in trade participation. We drop the xV subscripts to simplify notation and define a latent variable that is a function of the sales of the most productive firm and sales at the cutoff:

$$Z_{it} = \frac{\bar{r}_{it}}{r(c_{it}^U)} = \left(\frac{\bar{c}_{it}}{c_{it}^U} \right)^{1-\sigma}. \quad (\text{A.10})$$

If $Z_{it} > 1$, then at least one firm will export and trade is observed. However, trade may be observed even if $Z_{it} \leq 1$ because a firm survived from a prior export spell T periods ago when conditions were better. If a

firm survives with probability β^T , then it will continue to have positive export sales even if these sales are below what is implied by the cutoff, $r_{it}(c_v) < r(c_{it}^U)$. Alternatively, all firms from a previous period may be hit by the death shocks with probability $1 - \beta^T$. In that case, exit will occur but it will be unobservable to the econometrician if re-entry occurs, i.e. $Z_{it} > 1$.

We model entry and exit of product trade separately by conditioning on lagged trade participation. First, we take the log of equation (A.10) so that $z_{it} = \ln(Z_{it}) = (\sigma - 1) \ln \frac{c_{it}^U}{\bar{c}_{it}}$ is a latent variable that is a function of the cutoff condition and the minimum unit cost draw from the cohort $\ln \bar{c}_{it}$. Second, we let $\mathbf{1}_{it}^{Trade} = 1$ if trade is observed. The latter always occurs when $z_{it} > 0$. But we also observe trade if firms survive from a previous export spell, even if they are below the cutoff, or if all firms exit and at least one re-enters in the same period. So we must track lagged trade participation in period $t - 12$ as well. Rather than include a lagged dependent variable as a regressor, we use it to condition the sample. This method is sometimes called a restricted transition model.

If we condition on a sample of products that were non-traded in $t - 12$ then we can model the probability of entry as

$$\Pr(\mathbf{1}_{it}^{Trade} = 1 \mid \mathbf{1}_{it-12}^{Trade} = 0) = Probit[z_{it}] \quad (\text{A.11})$$

where *Probit* is a standard unit normal distribution with index function z_{it} . We know in this case that no firm was exporting in the previous period, so we do not need to consider deterioration relative to previous conditions that include a legacy of surviving firms. We can then estimate parameters on observable determinants of z_{it} that relate to the cutoff level.

For exit, we condition on the sample of traded products in $t - 12$. If the export capital of at least one firm survives until time t , then a product will continue to be traded regardless of business conditions. But there is another possibility that all legacy firms from a previous spell exogenously exit with probability $1 - \beta^T$. Even in this case, we may still observe continuing export participation in the product-level data. If at least one firm is productive enough to *re-enter* after making a sunk investment, i.e. $z_{it} > 0$, then we won't observe exit. The latter could occur if business conditions have not deteriorated at all or if they have deteriorated but there is still some subset of firms that is below the cost cutoff.

An observable exit only occurs in the event of death to the export capital of all firms exporting a product. The probability of exit is therefore:

$$\Pr(\mathbf{1}_{it}^{Trade} = 0 \mid \mathbf{1}_{it-12}^{Trade} = 1) = Probit[-(1 - \beta^T)z_{it}]. \quad (\text{A.12})$$

The index function of the exit probability has the opposite sign from entry and it is re-scaled downward in magnitude by $(1 - \beta^T)$. We ultimately include the same regressors on the RHS of the exit regression equations. So all coefficients should have the opposite sign of the entry equation (and the export value equation). The coefficients and marginal effects are adjusted downward relative to the entry coefficients to a degree that declines in the rate of survival β^T .

A.6 Measurement Error in Policy Risks

The trade policy tail risk factors may contain errors relative to the true exporter beliefs of the policy barriers that could prevail following Brexit. This is clear for the non-cooperative tariffs, which rely on inverse export supply elasticities. But it could also be the case for MFN tariffs. While the EU would mostly likely apply

its extant MFN tariff schedule to UK imports if no deal is reached, the UK may negotiate new commitments with WTO that are different from the current EU levels.

We use a multiple indicators method to address the measurement error. If exporters did not expect the MFN scenario to be exactly the current EU common external tariff, then we measure their beliefs with error. Suppose τ_V^M is the true tariff level exporters believe would prevail in a hard Brexit, but we measure the EU tariff, $\tau_V^{M,EU}$. We can relate the measured and true risk factors by $(1 - (\tau_V^{M,EU})^{-\sigma}) = (1 - (\tau_V^M)^{-\sigma}) + e_V^{EU}$, where e_V^{EU} is classic measurement error.

To address this measurement error we compute the median MFN tariff $\tilde{\tau}_V^M$ of the U.S., Canada, Japan and Australia for each 6-digit HS industry code, assuming that $(1 - (\tilde{\tau}_V^M)^{-\sigma}) = (1 - (\tau_V^M)^{-\sigma}) + \tilde{e}_V$. While both measures are correlated with the true belief, we assume the measurement errors are uncorrelated, $\text{Cov}(e_V^{EU}, \tilde{e}_V) = 0$. We interact our alternative measure $(1 - (\tilde{\tau}_V^M)^{-\sigma})$ with mbv_t . We then include these as instrumental variables for $(1 - (\tau_V^M)^{-\sigma})$, their interactions, and their lags that could be measured with error. We use the same approach and exogenous set of countries to construct instruments for the non-cooperative tariffs.

In the main text we report the overall weak identification F-stats of the full set of instrument variables. These statistics are inflated by the fact that a product tariff appears multiple times for every time period and country-product trade flow. In Table A3 we report F-statistics and coefficients of the zero-th stage regression at the highest level of aggregation for our policy data: the HS 6 digit product level. The F-statistic on the MFN and non-cooperative tariff estimates are above the threshold to rule out weak instruments concerns. Moreover, the coefficients on the instruments are positive and significant, which is sensible given their construction. We also note that the predicted regressors do not require an adjustment to standard errors because the IV procedure already handles this issue. Further discussion of measurement error correction in panel data is found in Griliches and Hausman (1986).

A.7 Contract Price Determinants

In this section, we explore the determinants of the contract price used in the econometric analysis to capture the probability of Brexit. In order to do so, we exploit the daily variation of the contract price and polling data along key dates in the year leading up to the referendum.

In Table A2 we present the results of different specifications to explain the contract price. In column 1 the results shows that the average daily ln contract price over this period is -1.21 when we include only a constant in the regression. One determinant is the ln share of exit voters. We add this variable to the specification in column 2 along with an indicator for the pre-wording period: July and August 2015 for which there is no opinion polls data and where we impute it with a constant equal to the September value. The marginal effect of the share of exit voters on the contract price in column 2 is positive and significant and this model explains almost 0.3 of the variation in the contract price.

In column 3 we add an indicator for the period in which the Brexit bill gained Royal Assent (December 17, 2015) and became the European Union Referendum Act of 2015. We interpret this as a key institutional step towards the actual possibility of a trade policy event. The results show an increase in both the average contract price as well as its elasticity with respect to exit poll in this period.

In Figure A1 we plot the predicted value of the contract obtained from the specification in column 3 and

averaged at the monthly level against its actual value. This is the variation that helps identify the first stage of the regressions when we address measurement error in the contract price using polling data.

A.8 Sunk Cost Industries Estimation

Only industries with positive sunk costs to export are affected by trade policy uncertainty. To identify these industries, we follow Handley and Limão (2017). We estimate the persistence of export participation at the product level conditional on a set of controls. Significant correlation of lagged and current participation can be interpreted as evidence of sunk costs.

We use UK HS-8 product import data from 2012-2016 from the non-EU OECD exporters plus Brazil, Russia, India, and China. This country set minimizes the risk of endogeneity that may arise from the EU countries over the period before the referendum. We define a variety $\tilde{V}$ as a HS-8 product observation and the corresponding HS-4 product code as an industry V . This yields 1,217 industries with an average of 7.7 varieties per industry and a standard deviation of 10.5. We estimate the model at the semester-level frequency. While we have higher frequency monthly data, our interest is in identifying longer run persistence in exporting. The persistence or lack thereof in monthly data might be driven by exogenous short-lived shocks or noise.

We estimate the following equation for the period ranging from the first semester of 2013 to the first semester 2016 for each V :

$$T_{\tilde{V}xt} = b_V^{sunk} \max \{T_{\tilde{V}xt-1}, T_{\tilde{V}xt-2}\} + b_{V,2012} T_{\tilde{V}x2012} + \alpha_{Vxt} + \alpha_{\tilde{V}} + \eta_{\tilde{V}xt}.$$

The dependent variable $T_{\tilde{V}xt}$ is an indicator taking the value of one if product $\tilde{V}$ is exported from country x at semester t to the UK. We control for initial conditions by $T_{\tilde{V}x2012}$, which is an indicator taking the value of one if product $\tilde{V}$ is exported by x at semester t to the UK in 2012. The terms α_{Vxt} and $\alpha_{\tilde{V}}$ are fixed effects capturing demand conditions, tariffs and transport costs, and industry conditions respectively. We interpret positive and significant estimates of b_V^{sunk} as evidence of sunk costs.

We estimate 1,032 of a possible 1,217 coefficients. We define as sunk cost (SC) industries those where the estimated coefficient has a t-statistic greater than 1.96. The t-statistic criterion provides a natural rank ordering of industries that is superior to using the estimated level of the persistence coefficient. In some cases, the estimated coefficients may be high, but they are imprecisely estimated. There are 809 industries we classify as high sunk cost (78%).

In Figure A2, we show the relation between the persistence coefficient and the t-statistic where the red line denotes a t-stat value of 1.96. It shows the same pattern as in Handley and Limão (2017) with all the significant coefficients between 0 and 1, as expected. The average significant persistence coefficient is 0.33 with a standard deviation of 0.11.

A.9 Identification of Average and MFN Risk Uncertainty Effects

In order to characterize the variation we employ to identify both the MFN and average risk effects under trade policy uncertainty, let's start with time-demeaning equation (15):

$$\ln \widetilde{R_{ixVt}} = \beta_{ix} \rho_{ix} \widetilde{mbv_t} r_{ixV}^s + \tilde{\alpha}_{xt} + \tilde{\alpha}_{it} + \tilde{\mathbf{e}}_{ixVt} \quad (\text{A.13})$$

where the “ $\widetilde{}$ ” notation indicates time-demeaned variables. The coefficients are $\beta_{ix} = \sum_{s=M,W} \sum_{l=0}^L W_{ix}^s(l)$ and we use the notation $r_{ixV}^s \equiv [1 - (\tau_{ixV}^s)^{-\sigma}]$ as a shorthand for policy risk factors. We also define $\rho_{ix} = 1$ if exports from x to a destination i face potential time varying risk and 0 otherwise. In our baseline specification $\rho_{ix} = 1$ for all bilateral trade flows given that we consider only UK-EU trade flows. In section 6.6 we consider third countries for which we assume no trade policy uncertainty, so in those cases we have $\rho_{ix} = 0$. In the following equation we rearrange equation A.13 to get the differential MFN risk:³

$$\ln \widetilde{R_{ixVt}} = \rho_{ix} \left[\beta_{ix}^A \widetilde{mbv_t} + \beta_{ix}^M \widetilde{mbv_t} \overline{r_{ixV}^M} \right] + \tilde{\alpha}_{xt}^X + \tilde{\alpha}_{it}^I + \tilde{\mathbf{e}}_{ixVt} \quad (\text{A.14})$$

where $\overline{r_{ixV}^M} \equiv r_{ixV}^M - \Sigma_V r_{ixV}^M / T_{V_{ix}}$, $T_{V_{ix}}$ is the number of industries for which there are products traded between this pair of countries, $\beta_{ix}^A = \sum_{s=M,W} W_{ix}^s \Sigma_V r_{ixV}^M / T_{V_{ix}}$ is the average effect, and $\beta_{ix}^M = W_{ix}^M$ is the differential MFN risk effect.

Baseline. In our baseline specification we cannot identify the average uncertainty effect given that it is perfectly captured by fixed effects. Note that we only use EU-UK trade and therefore we can write $\tilde{\alpha}_{ixt} = \tilde{\alpha}_{xt}^X + \tilde{\alpha}_{it}^I$. Given that $\rho_{ix} = 1$ for all flows and $r_{ixV}^M = r_V^M$, taking differences across industries within exporter-importer-month yields:

$$\ln \widetilde{R_{ixVt}} - \ln \widetilde{R_{ixV't}} = \beta^M \widetilde{mbv_t} \left\{ \overline{r_V^M} - \overline{r_{V'}^M} \right\} + \tilde{\mathbf{e}}_{ixVt} - \tilde{\mathbf{e}}_{ixV't} \quad (\text{A.15})$$

Equation A.15 shows the sources of variation we use to identify the MFN risk uncertainty effect in the baseline: differences in probability over time ($\widetilde{mbv_t}$), and differences in MFN tail risk across industries ($\overline{r_V^M} - \overline{r_{V'}^M}$).

Average Uncertainty Effect. In order to identify the average effect, we add within EU exports and a set of non-EU countries (ROW) to our estimating sample. We assume that there is no trade policy uncertainty in flows that do not involve trade between UK and EU countries. Therefore, we assume the following treatment status: $\rho_{uk,eu} = 1$, $\rho_{eu,uk} = 1$, $\rho_{eu,eu'} = 0$, $\rho_{row,eu} = 0$, $\rho_{row,uk} = 0$, $\rho_{eu,row} = 0$, $\rho_{uk,row} = 0$.

We can start from equation A.14 and take the first difference for a given EU importer-industry-month between UK and a another exporter not facing uncertainty in the EU. Given the diff-in-diff variation we employ to identify the average uncertainty effect, we need to include exporters that do not face uncertainty in countries where UK does not face uncertainty either. Eurostat only provides trade between EU countries and between EU and ROW countries, which implies we have to employ intra-EU trade where UK is not

³Without loss of generality, we model the probability with a single lag in this section.

involved and assume there is no uncertainty in those flows.

$$\ln \widetilde{R_{eu,uk,V,t}} - \ln \widetilde{R_{eu,eu',V,t}} = \beta^A \widetilde{mbv}_t + [\tilde{\alpha}_{uk,t} - \tilde{\alpha}_{eu',t}] + \tilde{\mathbf{e}}_{eu,uk,V,t} - \tilde{\mathbf{e}}_{eu,eu',V,t} \quad (\text{A.16})$$

The second difference needs to eliminate $[\tilde{\alpha}_{uk,t} - \tilde{\alpha}_{row,t}]$, the difference in the exporter fixed effects. Here is when we employ ROW data and assume that exports from UK and EU to these countries faced no trade policy uncertainty.⁴ Hence, we can take the second difference by using equation A.16 with ROW as an importer:

$$\begin{aligned} \left[\ln \widetilde{R_{eu,uk,V,t}} - \ln \widetilde{R_{eu,eu',V,t}} \right] - \left[\ln \widetilde{R_{row,uk,V,t}} - \ln \widetilde{R_{row,eu',V,t}} \right] = \\ \beta^A \widetilde{mbv}_t + \left[\tilde{\mathbf{e}}_{eu,uk,V,t} - \tilde{\mathbf{e}}_{eu,eu',V,t} \right] - \left[\tilde{\mathbf{e}}_{row,uk,V,t} - \tilde{\mathbf{e}}_{row,eu',V,t} \right] \end{aligned} \quad (\text{A.17})$$

Equation A.17 shows that we use diff-in-diff variation to identify the average uncertainty effect by using intra-EU and EU-ROW trade for each industry-month flow. Note that in this example we are comparing trade flows for a given industry-month to stress the source of variation used in identifying the average effect. However, we are also able to identify β^M when we include third countries as in the baseline for two reasons: (1) we also compare different industry-months across these differences; and (2) we use within exporter-importer-month industry variation, as in the baseline.

A.10 Heterogeneous Exporter Elasticities

We provide evidence that the heterogeneity in elasticities in the baseline IV of Table 10, replicated in columns 1 and 2 of Table A10, is partially explained by differential beliefs related to policy uncertainty and exacerbated after we control for omitted bilateral-industry trends and composition.

- Measurement error in probability of Brexit. In columns 3-8 address the possibility that measurement error in the probability was differentially correlated with EU exports over this period and thus generate heterogeneity. One specific reason would be that UK exporters pay closer attention to the probability of Brexit so that any measurement error in it would cause larger attenuation for $\mathcal{E}_{UK}^M$.
 - In columns 3, 4 we address the measurement error by instrumenting mbv_t with polling data as described for the homogeneous coefficients case. Both elasticity estimates increase, consistent with attenuation bias, and the one for the UK does so by more, which is consistent with UK exporters paying closer attention to this probability.
 - In columns 5 and 6 we show the heterogeneous effect remains if we use a transaction weighted average of the prediction within month.
 - Columns 7 and 8 indicate less heterogeneity when we allow the regression to place higher weight on months with higher prediction market transactions, which may reflect more accurate information. We find $\mathcal{E}_{EU}^M / \mathcal{E}_{UK}^M = 1.35$ due to a relatively larger increase in $\mathcal{E}_{UK}^M$, which is consistent with UK exporters paying more attention to this measure. We can reject equality of coefficients only at a p-value of 0.18

- We tested for composition effects in two distinct ways, both of which increase the heterogeneity.

⁴We do not include countries that have PTAs with EU and may potentially face uncertainty from Brexit.

- Controlling for bilateral-hs6 trends, e.g. if EU exports to UK in risky products had been falling by more for some other reason in this period then the uncertainty coefficient for the EU would be higher. In columns 9 and 10 of Table A10 we find that this is not the case. Both the EU and UK elasticities are lower relative to the baseline but their ratio is now higher, at about 2.6.
- Limiting the sample to industries where the elasticity of substitution, σ , are more similar: specifically between 2 and 6. The rationale is that if the EU countries tended to export a larger fraction of products with higher σ and we assume it is common to all (to construct risk) then this could induce a higher average uncertainty elasticity. In columns 11-12 we find the heterogeneity in the baseline coefficient increases and we reject their equality.

A.11 Predicted and Estimated Post-Referendum Annual Change in Exports

Predicted Changes in Exports. To quantify the impact of Brexit we can employ the cross-elasticity $|\mathcal{E}^M|$ estimated in the pre-referendum period, and infer the out-of-sample predicted annual change in exports:

$$\widehat{\ln R} = \widehat{\mathcal{E}^M} \times \overline{(1 - (\tau^M)^{-\sigma})} \times [mbv_{post} - mbv_{pre}] \quad (\text{A.18})$$

where $\overline{(1 - (\tau^M)^{-\sigma})}$ is the average MFN risk, mbv_{pre} is the observed (ln) contract price before the referendum, and mbv_{post} is the (unobserved) post-referendum uncertainty. As explained in the text, we use the unstandardised coefficient from Table 6, column 5 for $\widehat{\mathcal{E}^M}$ (-1.29) and the transaction-weighted average pre-referendum mbv (-1.21). Given that we do not observe a direct proxy for the probability after the referendum, we assume that $mbv_{post} = 0$ for that period of time (which implies a contract price equal to 1), and also assume that there is some “Bregret” after the vote as we argue in the text.

We are making an out-of-sample prediction knowing that the Brexit option won the referendum. In the baseline period, we assume that the probability of drawing a policy from H^{BR} conditional on the Brexit voting outcome in the referendum is fixed, as given by p^{BR} in (9). Thus we can write $\ln m_t/m_0 = mbv_t - mbv_0$ for the baseline estimation. Relative to our estimation, any change in the post-referendum probability of Brexit are captured by changes to p^{BR} —the parameter we relate to **Bregret**. Given a change in p^{BR} , we now write the change in the probability as $\ln[m_t/m_0] = [\ln p_{post}^{BR} + mbv_{post}] - [\ln p_{pre}^{BR} + mbv_{pre}]$. Noting that $mbv_{post} = 0$, we can use equation (20) to get:

$$\widehat{\ln R} = \widehat{\mathcal{E}^M} \times \overline{(1 - (\tau^M)^{-\sigma})} \times [-mbv_{pre}](1 - Bregret) \quad (\text{A.19})$$

Therefore, we have to assume a value for *Bregret*. In the paper, we argue that a plausible value is 1/3, which delivers $\widehat{\ln R} = -0.156$.

Estimated Changes in Exports. We can compare the out-of-sample prediction with the estimated change in exports shown in Table 13, column 3. We derive the empirical equation as follows. First, note that taking yearly differences yields:

$$\Delta_T \ln R_{ixVt} = \sum_{l=0}^L W_{ix}^M(l) \left\{ (mbv_{t-l} - mbv_{t-l-12})(1 - Bregret) \left[1 - (\tau_{ixV}^M)^{-\sigma} \right] \right\} + \alpha_{it,xt} + \epsilon_{ixVt} \quad (\text{A.20})$$

where Δ_T captures the annual change $\ln R_{ixV,t} - \ln R_{ixV,t-12}$, and $t \in post$ and $t-12 \in pre$ in the September

2016-June 2017 sample we employ. We then assume that the post-referendum mbv is fixed, $mbv_{t-l} = \overline{mbv}$, and potentially zero. This yields the following estimating equation:

$$\Delta_T \ln R_{ixVt} = \beta^{post}(-mbv_{t-l-12}) \left[1 - (\tau_{ixV}^M)^{-\sigma} \right] + \alpha_{ixV,it,xt} + \epsilon_{ixVt} \quad (\text{A.21})$$

where $\beta^{post} = \sum_{l=0}^L W_{ix}^M(l) (1 - Bregret)$, and we include ixV fixed effects to control for the interaction $\overline{mbv} \left[1 - (\tau_{ixV}^M)^{-\sigma} \right]$ in case $\overline{mbv} < 0$, and potential ixV linear trends.

When we estimate this equation we get $\widehat{\beta^{post}} = -0.531$, which is the estimate we use along with the average pre-referendum mbv and average MFN risk to calculate the estimated change in Table 13, column 3 ($-0.531 \times 1.21 \times 0.149 = -0.096$). Note that $\widehat{\beta^{post}}$ includes any Bregret by construction and therefore it is expected to be lower than the baseline estimate in which we only use the pre-referendum period (-1.29).

Measurement error in post-referendum mbv . The derivation in the previous section assumes a fixed mbv in the post-referendum period. However, if the actual information it is capturing varies over time then we may be introducing measurement error that attenuates the cross-elasticity estimate. Assume that the actual post-referendum value is $mbv_t^{post} = \overline{mbv} + \kappa_t$. Then the correct specification would be as follows:

$$\Delta_T \ln R_{ixVt} = \beta^{post}(\kappa_t - mbv_{t-l-12}) \left[1 - (\tau_{ixV}^M)^{-\sigma} \right] + \alpha_{ixV,it,xt} + \epsilon_{ixVt} \quad (\text{A.22})$$

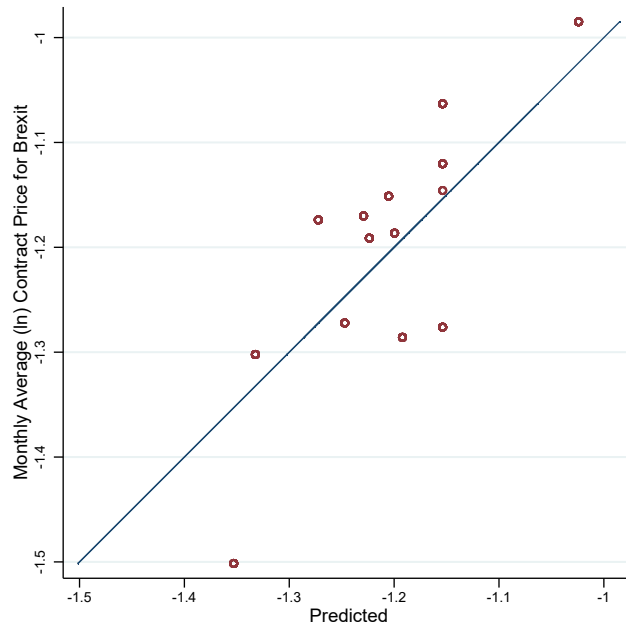
where note that $\overline{mbv}$ will be absorbed by the ixV fixed effect as before. Therefore, we are estimating the following equation:

$$\Delta_T \ln R_{ixVt} = \beta^{post}(-mbv_{t-l-12}) \left[1 - (\tau_{ixV}^M)^{-\sigma} \right] + \alpha_{ixV,it,xt} + \left[\epsilon_{ixVt} + \beta^{post} \kappa_t \left[1 - (\tau_{ixV}^M)^{-\sigma} \right] \right] \quad (\text{A.23})$$

Equation A.23 shows that if $COV(mbv_{t-12} \left[1 - (\tau_{ixV}^M)^{-\sigma} \right], \kappa_t \left[1 - (\tau_{ixV}^M)^{-\sigma} \right]) > 0$, then the estimate for β^{post} will be attenuated. This may be the case if the Brexit sentiment has a seasonal factor that is captured by both mbv_{t-12} and κ_t , given that both t and $t - 12$ correspond to the same month.

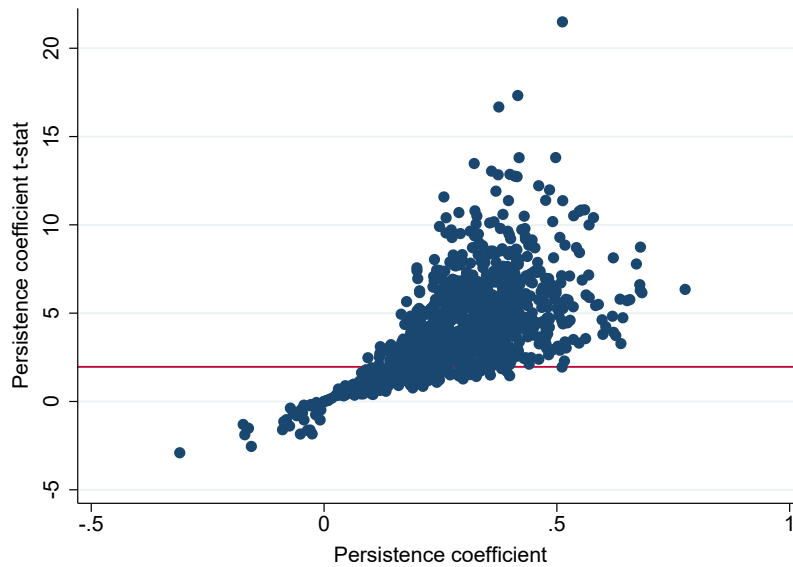
The lower coefficient we observe when we use post-referendum data to estimate the coefficient may be both due to a change in the conditional probability of Brexit (Bregret) and measurement error coming from the fact that we do not observe post-referendum contract prices. In fact, the difference between the point estimate of the predicted export value change in Table 13, column 1 and the one implied by using post-referendum data in Table 13, column 3 suggest that both are plausible.

Figure A1: Brexit Referendum Contract vs. Predicted Beliefs



Notes: Average monthly contract price used in the baseline sample. The average of predicted monthly contract is based on the daily model in Table A2 including the exit share, date bill passed and interactions.

Figure A2: Sunk Cost Estimates. t -statistics vs Persistence Coefficient.



Notes: t -statistics on persistence coefficient by HS-4 industry headings relative to estimated persistence coefficient. Red line is a t -statistic of 1.96.

Table A1: MFN Risk by Sector - UK and EU Trade

Sector	Percentage of Total Exports (2015)	MFN Risk		
		Mean	Median	SD
1 Animals	0.7	0.271	0.317	0.15
2 Vegetables	1.5	0.16	0.145	0.152
3 Fats & Oils	0.2	0.182	0.21	0.103
4 Prepared Foodstuffs	2.4	0.341	0.382	0.222
5 Minerals	7.2	0.0136	0	0.0278
6 Chemicals	16.4	0.16	0.193	0.0778
7 Plastics, Rubber & Articles	5.3	0.164	0.217	0.0791
8 Hides, Leather, & Articles	0.6	0.127	0.129	0.0749
9 Wood, Straw & Articles	0.8	0.0896	0.0613	0.0984
10 Pulp, Paper & Articles	2.3	0	0	0
11 Textiles & Articles	3.4	0.271	0.265	0.0881
12 Footwear, Headgear, other	1.0	0.230	0.168	0.149
13 Stone, Plaster, Cement, other	1.0	0.127	0.112	0.0977
14 Precious stones, Metals, Jewellery	1.8	0.0289	0	0.0508
15 Base Metals & Articles	5.3	0.0732	0.0652	0.0769
16 Machinery; Elec. Equip.; Electronics	22.4	0.0735	0.0652	0.0621
17 Vehicles, Aircraft, Vessels	21.8	0.154	0.135	0.116
18 Optical, Medical & other instruments	3.6	0.0856	0.100	0.0597
19 Arms and Ammunition	0.0	0.0997	0.101	0.0171
20 Miscellaneous Manufactures	2.3	0.0968	0.101	0.0583
21 Art and Antiques	0.1	0	0	0
Overall	100	0.145	0.123	0.125

Notes: EU-UK subsample. Exporter percent shares calculated for 2015. MFN risk defined as $1-(\tau^{\text{MFN}})^{-\sigma}$, where $\sigma=4$ and $\tau^{\text{MFN}}=1+\text{MFN advalorem}/100$. HS6 codes within each section included if present in the baseline sample of continuing country-product HS-6 varieties for 8/15-6/16.

Table A2: Market Based Variable Daily Determinants

Daily Contract Price (ln) 5/27/15 - 6/22/16

	1	2	3
Share of Exit Voters (ln)		2.69 (0.231)	2.26 (0.484)
Bill Passed (BP)			0.808 (0.422)
BP×Exit Voters (ln)			1.14 (0.547)
Pre-Wording		0.101 (0.0162)	0.0699 (0.0192)
Constant	-1.21 (0.00794)	0.815 (0.175)	0.513 (0.376)
N	393	393	393
R2	0.000	0.290	0.325

Notes: OLS regressions. Contract price is the ln daily average price defined between 0 and 1 of the Brexit shares in predicitt.com (pays \$1 if the Brexit option wins the referendum), share of exit is the daily log share of those that responded they would vote to exit (adjusted by 1 - undecided) and is based on different opinion polls constructed by Number Cruncher Politics (shares between 0 and 1), Bill Passed is an indicator that takes the value of one after the European Union Referendum Bill was given Royal Assent on 12/17/15, Pre-Wording is an indicator that takes the value of one before 9/1/15, when the final wording of the referendum question was established. Share of exit is not available before 9/1/15 and is imputed to 9/15 monthly average. Robust standard errors in parenthesis.

Table A3: MFN and Trade War Risk

	1	2	3
	OLS	OLS	OLS
<i>Dependent Variable:</i>	MFN risk	MFN risk	Trade War risk
Median MFN risk	0.832 (0.0147)	0.820 (0.016)	-0.050 (0.0342)
Median Trade War risk	.	0.007 (0.0065)	0.114 (0.0137)
Constant	0.078 (0.00206)	0.0781 (0.00452)	0.656 (0.00998)
N	4,128	3,390	3,390
R2	0.387	0.383	0.0233
F-statistic	3192	1311	36.4

Notes: OLS regressions. Columns 1: MFN risk regressed on median HS6-specific MFN risk across four large countries (Australia, Canada, Japan and US). MFN risk defined as $1 - (\tau \text{MFN}) - \sigma$, where $\sigma = 4$ and $\tau \text{MFN} = 1 + \text{MFN advalorem}/100$. Columns 2 and 3: MFN risk and Trade War risk regressed on median HS6-specific MFN risk and Trade War defined across the same four large countries. Only HS6 products traded between EU and UK over the baseline sample included. Column 1 corresponds to IV results in all tables other than column 4 of Table 2. Robust standard errors in parenthesis.

Table A4: UK and EU Alternative Brexit Measures and Entry/Exit
Annual Product Entry and Exit Indicators, 8/15-6/16 relative to 8/14-6/15

	<i>Entry</i>			<i>Exit</i>		
	1	2	3	4	5	6
	Prediction Probability		Polls	Prediction Probability		Polls
	Unweighted Average	Contract Weighted Average		Unweighted Average	Contract Weighted Average	
Pr(Brexit)×MFN Risk	-0.606 (0.0898)	-0.589 (0.0854)	-2.21 (0.398)	0.204 (0.0375)	0.192 (0.0358)	0.482 (0.164)
N	647,488	647,488	647,488	977,177	977,177	977,177

Notes: IV regressions. Entry sample is defined as exporter-importer-HS6 products not traded in the same month of the previous year, and exit sample is defined as exporter-importer-HS6 products traded in the same month of the previous year. Dependent variable $Entry(t)=1$ if $Export(t)>0$ and $Export(t-12)=0$ for the exporter-importer-HS6 observation in month t from 8/15-6/16. $Exit(t)=1$ if $Export(t)=0$ and $Export(t-12)>0$ for the exporter-importer-HS6 observation in month t from 8/15-6/16. $Pr(Brexit)$ defined as the monthly average (ln) leave prediction market contract price, and MFN risk defined as $1-(\tau^{MFN})^{-\sigma}$, where $\sigma=4$ and $\tau^{MFN}=1+MFN \text{ advalorem}/100$. Columns 1 and 4 use the baseline unweighted monthly average of daily $Pr(brexit)$, columns 2 and 5 weight the daily probabilities with the squared root of the volume of daily transactions. Columns 3 and 6 define $Pr(Brexit)$ as the average (ln) share of exit voters adjusted by 1 - undecided. The poll data for July and August 2015 is imputed to be the same as in September when polling begins. Coefficients report the sum of the included lags. Robust standard errors clustered at the exporter-importer-HS6 level in parenthesis. All estimations include exporter-importer-HS6, exporter-month and importer-month fixed effects.

Table A5: UK and EU Alternative MFN Risk Measures and Entry/Exit
Annual Product Entry and Exit Indicators, 8/15-6/16 relative to 8/14-6/15

	Entry			Exit				
	1	2	3	4	5	6	7	8
	$\sigma=2$	$\sigma=3$	Industries with $\sigma \in [2, 6]$	(ln) MFN tariff threat	$\sigma=2$	$\sigma=3$	Industries with $\sigma \in [2, 6]$	(ln) MFN tariff threat
Pr(Brexit) $\times$ MFN Risk	-1.13 (0.162)	-0.781 (0.114)	-0.703 (0.114)	-2.09 (0.292)	0.376 (0.0666)	0.261 (0.0472)	0.183 (0.0439)	0.688 (0.118)
N	647,488	647,488	344,915	647,488	977,177	977,177	555,850	977,177

Notes: IV regressions. Entry sample is defined as exporter-importer-HS6 products not traded in the same month of the previous year, and exit sample is defined as exporter-importer-HS6 products traded in the same month of the previous year. Dependent variable Entry(t)=1 if Export(t)>0 and Export(t-12)=0 for the exporter-importer-HS6 observation in month t from 8/15-6/16. Exit(t)=1 if Export(t)=0 and Export(t-12)>0 for the exporter-importer-HS6 observation in month t from 8/15-6/16. Pr(Brexit) defined as the monthly average (ln) leave prediction market contract price and MFN risk defined as $1 - (\tau^{\text{MFN}})^{\sigma}$, where $\sigma=4$ in columns 5-6, $\tau^{\text{MFN}} = 1 + \text{MFN advalorem}/100$. Columns 1 and 5 use $\sigma=2$, columns 2 and 6 use $\sigma=3$ and columns 3 and 7 exclude industries with σ higher than 6 and lower than 2 based on estimations using US import data in Broda and Weinstein (2006). Columns 4 and 8 use (ln) τ^{MFN} as the MFN risk measure. Coefficients report the sum of current and two monthly lags. Robust standard errors clustered at the exporter-importer-HS6 level in parenthesis. All estimations include exporter-importer-HS6, exporter-month and importer-month fixed effects.

Table A6: UK and EU MFN Risk - Robustness to Unobserved Shocks

Monthly Export Value (ln) 8/15-6/16				
	1	2	3	4
<i>Extra FE relative to Baseline</i>	Section-month	Section-month-bilateral	Section trend-bilateral	HS6 trend-bilateral
Pr(Brexit)×MFN Risk	-2.16 (0.423)	-2.03 (0.430)	-1.31 (0.198)	-1.20 (0.207)
N	637,263	636,339	637,263	637,263

Notes: IV regressions. Dependent variable ln(exports) defined at the exporter-importer-HS6-month level for UK and EU (2015 membership). Pr(Brexit) defined as the monthly average (ln) leave prediction market contract price, and MFN risk defined as $1 - (\tau^{MFN})^{-\sigma}$, where $\sigma=4$ and $\tau^{MFN} = 1 + \text{MFN advalorem}/100$. We instrument the MFN risk by the median HS6-specific MFN risk across four large countries (Australia, Canada, Japan and US). Coefficients report the sum of current and two monthly lags. Robust standard errors clustered at the exporter-importer-HS6 level in parenthesis. All include exporter-importer-HS6 and exporter-month, importer-month effects. The alternative unobserved shocks are captured by the fixed effects noted in each column: section-month (1) section-month-bilateral (2), section linear trend-month-bilateral (3), HS6 linear trend-month-bilateral (4). We define sectors using the 21 sections of the HS that group related HS-6 digit codes.

Table A7: UK and EU MFN Risk - Change in Export Values Relative to Pre-Referendum Periods

Monthly Export Value Change ($\Delta \ln$)—8/15-6/16, Relative to Same Month in Previous Period				
	1	2	3	4
<i>Relative to:</i>	2012	2013	2014	2015
Pr(Brexit)×MFN Risk	-0.601 (0.282)	-0.544 (0.272)	-0.701 (0.262)	-0.581 (0.247)
N	577,495	595,770	609,397	625,278

Notes: IV regressions. Dependent variable is change in ln(exports) defined at the exporter-importer-HS6-month level for UK and EU (2015 membership) relative to same month in time period that ends in June of the year in the column headings. Samples includes all exporter-importer-HS6 trade flows from the baseline levels regression for which a log change can be computed relative to prior time period. Pr(Brexit) defined as the monthly weighted average (ln) leave prediction market contract price using the square root of daily volume as the weight. MFN risk defined as $1 - (\tau^{MFN})^{-\sigma}$, where $\sigma=4$ and $\tau^{MFN} = 1 + \text{MFN advalorem}/100$. All columns instrument the risk variables by their respective median HS6-specific risk across four large countries (Australia, Canada, Japan and US). Coefficients report the sum of current and two monthly lags. Robust standard errors clustered at the exporter-importer-HS6 level in parenthesis. All estimations include exporter-importer-HS6, exporter-month and importer-month fixed effects.

Table A8: UK and EU MFN Risk - Export Values: Restricted Subsamples of Baseline Based on Continuous Trade

Monthly Export Value (ln) 8/15-6/16				
	1	2	3	4
	Baseline	Also traded in		
	subsample	2014	2014, 13	2014, 13, 12
Pr(Brexit) $\times$ MFN Risk	-1.45 (0.198)	-1.31 (0.197)	-1.24 (0.199)	-1.22 (0.198)
Obs	637,263	602,031	577,114	547,279
Share obs	1	0.945	0.906	0.859
Share trade	1	0.988	0.977	0.956

Notes: IV regressions. Dependent variable $\ln(\text{exports})$ defined at the exporter-importer-HS6-month level for UK and EU (2015 membership). Pr(Brexit) defined as the monthly average (ln) leave prediction market contract price, and MFN risk defined as $1 - (\tau^{\text{MFN}})^{-\sigma}$, where $\sigma=4$ and $\tau^{\text{MFN}} = 1 + \text{MFN advalorem}/100$. We instrument the risk variables by their respective median HS6-specific risk across four large countries (Australia, Canada, Japan and US). Coefficients report the sum of current and two monthly lags. Robust standard errors clustered at the exporter-importer-HS6 level in parenthesis. All estimations include exporter-importer-HS6, exporter-month and importer-month fixed effects.

Table A9: Heterogeneous UK and EU MFN and Average Risk - Alternative Specifications

Monthly Export Value (ln) 8/15-6/16						
<i>Sample (rel. to baseline):</i>	1	2	3	4	5	6
	OLS	OLS	IV	OLS	OLS	IV
Panel A: Average Effect						
Pr(Brexit) $\times$ UK-EU	-0.238 (0.0373)	-0.118 (0.0432)	-0.00525 (0.0480)	-0.188 (0.0419)	-0.0183 (0.0639)	0.130 (0.0761)
Pr(Brexit) $\times$ MFN Risk $\times$ UK-EU		-0.805 (0.148)	-1.56 (0.199)		-1.33 (0.374)	-2.49 (0.488)
Exporter-Importer-HS6 FE	Yes	Yes	Yes	Yes	Yes	Yes
Exporter-Month and Importer-Month FE	Yes	Yes	Yes	No	No	No
Exporter-HS6-Month and Importer-HS6-Month FE	No	No	No	Yes	Yes	Yes
N	7,447,011	7,447,011	7,447,011	6,869,478	6,869,478	6,869,478
R2	0.872	0.872	n/a	0.913	0.913	n/a
Panel B: Heterogeneous Effect						
UK:						
Pr(Brexit) $\times$ UK	-0.202 (0.052)	-0.100 (0.0595)	-0.0271 (0.0652)	-0.164 (0.059)	-0.00610 (0.0893)	0.0918 (0.105)
Pr(Brexit) $\times$ MFN Risk $\times$ UK		-0.685 (0.191)	-1.17 (0.255)		-1.27 (0.539)	-2.06 (0.694)
EU:						
Pr(Brexit) $\times$ EU	-0.280 (0.0517)	-0.136 (0.0619)	0.0336 (0.0705)	-0.215 (0.0574)	-0.0335 (0.0892)	0.174 (0.108)
Pr(Brexit) $\times$ MFN Risk $\times$ EU		-0.967 (0.235)	-2.10 (0.316)		-1.38 (0.509)	-2.95 (0.670)
Exporter-Importer-HS6 FE	Yes	Yes	Yes	Yes	Yes	Yes
Exporter-Month and Importer-Month FE	Yes	Yes	Yes	No	No	No
Exporter-HS6-Month and Importer-HS6-Month FE	No	No	No	Yes	Yes	Yes
N	7,447,011	7,447,011	7,447,011	6,869,478	6,869,478	6,869,478
R2	0.872	0.872	n/a	0.913	0.913	n/a

Notes: Dependent variable $\ln(\text{exports})$ defined at the exporter-importer-HS6-month level for UK and EU (2015 membership). Pr(Brexit) defined as the monthly average $\ln$ leave prediction market contract price, and MFN risk defined as $1 - (\tau^{\text{MFN}})^{\sigma}$, where $\sigma=4$ and $\tau^{\text{MFN}}=1+\text{MFN advalorem}/100$. UK-EU is a binary indicator for UK exports to the EU and EU exports to the UK, UK is a binary indicator for UK exports to EU and EU is a binary indicator for EU exports to UK. The sample includes all exports and imports of the UK and EU to the OECD (excluding countries that have a PTA with EU: Chile, Israel, Korea, Mexico and Turkey) and four large developing countries: Brazil, Russia, India and China (BRIC). Coefficients report the sum of current and two monthly lags. Robust standard errors clustered at the exporter-importer-HS6 level in parenthesis.

Table A10: UK and EU MFN Risk by Exporter - Robustness

Robustness		Monthly Export Value (ln) 8/15-6/16																							
		1		2		3		4		5		6		7		8		9		10		11		12	
		Baseline		Poll IV		Weighted contract		Weighted across months		HS6-bilateral time trends		Composition: $\sigma \in [2, 6]$													
Exporter:		UK	EU	UK	EU	UK	EU	UK	EU	UK	EU	UK	EU	UK	EU	UK	EU	UK	EU	UK	EU	UK	EU		
Pr(Brexit)×MFN Risk		-1.05 (0.253)	-2.01 (0.315)	-3.14 (0.341)	-3.26 (0.413)	-0.922 (0.242)	-1.82 (0.300)	-2.18 (0.359)	-2.94 (0.429)	-0.722 (0.264)	-1.88 (0.331)	-0.985 (0.303)	-2.49 (0.373)												
H0: Identical MFN cross-elasticity (p-value)		0.0174		0.821		0.0202		0.176		0.0286		0.00178													
EU/UK elasticity		1.91		1.04		1.97		1.35		2.60		2.53													
N		369,589	267,674	369,589	267,674	369,589	267,674	369,589	267,674	369,589	267,674	213,510	155,474												

Notes: IV regressions. Dependent variable $\ln(\text{exports})$ defined at the exporter-importer-HS6-month level for UK and individual EU 2015 members. The source of exports is labelled on top of the columns. Pr(Brexit) defined as the monthly average (ln) leave prediction market contract price, and MFN risk defined as $1 - (\tau^{\text{MFN}})^{\sigma}$, where $\sigma=4$ and $\tau^{\text{MFN}} = 1 + \text{MFN ad valorem}/100$. Coefficients report the sum of current and two monthly lags. Robust standard errors clustered at the exporter-importer-HS6 level in parenthesis. All estimations include exporter-month and importer-month fixed effects. Additional effects are added in the column labelled "HS6-bilateral time trends".

Table A11: UK and EU MFN Risk - Entry and Exit by Exporter
Annual Product Entry and Exit Indicators, 8/15-6/16 Relative to 8/14-6/15

	<i>Entry</i>		<i>Exit</i>	
	1	2	3	4
	IV		IV	
<i>Exporter:</i>	UK	EU	UK	EU
Pr(Brexit)×MFN Risk	-0.239 (0.125)	-1.01 (0.130)	0.153 (0.0489)	0.275 (0.0585)
N	373,946	273,542	584,559	392,618

Notes: IV regressions. Entry sample is defined as exporter-importer-HS6 products not traded in the same month of the previous year, and exit sample is defined as exporter-importer-HS6 products traded in the same month of the previous year. Entry(t)=1 if Export(t)>0 and Export(t-12)=0 for the exporter-importer-HS6 observation in month t from 8/15-6/16. Exit=1 if Export(t)=0 and Export(t-12)>0. Pr(Brexit) defined as the monthly average (ln) leave prediction market contract price, and MFN risk defined as $1-(\tau^{MFN})^{-\sigma}$, where $\sigma=4$ and $\tau^{MFN}=1+MFN$ advalorem/100. We instrument the MFN risk by the median HS6-specific MFN risk across four large countries (Australia, Canada, Japan and US). In columns 1 and 3 the sample is UK exports to EU, and in columns 2 and 4 the sample is EU exports to UK. Coefficients report the sum of current and two monthly lags. Robust standard errors clustered at the exporter-importer-HS6 level in parenthesis. All estimations include exporter-importer-HS6, exporter-month and importer-month fixed effects.

Table A12: UK and EU MFN Risk - Change in Exports Post-Pre Referendum Period
Change Monthly Export Value ($\Delta \ln$)—8/16-6/17 Relative to Same Month in Referendum Period

	1	2	3	4	5	6
	OLS	IV	OLS	IV	OLS	IV
	None		Within Month		Within and Across Months	
$\Delta \text{Pr(Brexit)} \times \text{MFN Risk}$	-0.46 (0.192)	-0.605 (0.252)	-0.416 (0.183)	-0.531 (0.240)	-0.519 (0.276)	-0.862 (0.364)
N	619,859	619,859	619,859	619,859	619,859	619,859
R2	0.890	n/a	0.890	n/a	0.929	n/a

Notes: IV regressions. Dependent variable is change in ln(exports) after the referendum defined at the exporter-importer-HS6-month level for UK and EU (2015 membership) relative to same month in the previous year. Samples includes all exporter-importer-HS6 trade flows from the baseline levels regression for which a log change can be computed. $\Delta \text{Pr(Brexit)}$ defined as change in the monthly average (ln) leave prediction market contract price from post referendum, when ln(contract price)=0, less the average in the same month from 8/15-6/16 leading up to the referendum. Columns 1-2 use simple mean within month. Columns 3-4 weight within month using the square root of daily volume. Columns 5-6 use the weighted monthly average and weight the regression by the sum of trading volume in each month. MFN risk defined as $1-(\tau^{MFN})^{-\sigma}$, where $\sigma=4$ and $\tau^{MFN}=1+MFN$ advalorem/100. All columns instrument the risk variables by their respective median HS6-specific risk across four large countries (Australia, Canada, Japan and US). Coefficients report the sum of current and two monthly lags. Robust standard errors clustered at the exporter-importer-HS6 level in parenthesis. All estimations include exporter-importer-HS6, exporter-month and importer-month fixed effects.

Table A13: UK and EU MFN Risk - Robustness to Alternative Entry/Exit Measurement

Annual Product Entry and Exit Indicators, 8/15-6/16 Relative to 8/14-6/15

	<i>Entry</i>			<i>Exit</i>		
	1	2	3	4	5	6
<i>Initial Status at Month t-</i>	12	12,13	12,13,14	12	12,13	12,13,14
<i>Final Status at Month t-</i>	0	0 or 1	0 or 1 or 2	0	0 or 1	0 or 1 or 2
Pr(Brexit)×MFN Risk	-0.606 (0.0898)	-0.960 (0.131)	-0.977 (0.166)	0.204 (0.0375)	0.244 (0.0418)	0.266 (0.0440)
N	647,488	513,817	436,997	977,177	848,246	789,415

Notes: IV regressions. Entry sample at t is defined as exporter-importer-HS6 products not traded in any t-n, where the number of months n is shown in "Initial status" row; the exit sample is defined as exporter-importer-HS6 products traded in all t-n. Dependent variable Entry(t)=1 if Export(t)>0 (column 1), either t or t-1 (column 2) or t, t-1, t-2 (column 3). Exit(t)=1 if Export(t)=0 (column 4) t or t-1 (column 5) and t,t-1,t-2 (column 6). Pr(Brexit) defined as the monthly average (ln) leave prediction market contract price, and MFN risk defined as $1 - (\tau^{MFN})^{-\sigma}$, where $\sigma=4$ and $\tau^{MFN}=1+MFN \text{ advalorem}/100$. All columns instrument the risk variables by their respective median HS6-specific risk across four large countries (Australia, Canada, Japan and US). Coefficients report the sum of the included lags. Robust standard errors clustered at the exporter-importer-HS6 level in parenthesis. All estimations include exporter-importer-HS6, exporter-month and importer-month fixed effects.

Reference

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